EE1010 Math for EE — Exam 2 — Nov 20, 2025

- There are 5 Questions for 40 marks.
- Write your answers neatly, legibly and logically. Keep your proofs thorough. Your proofs must follow from the basics discussed in class. You may use any result from class directly with mention (without re-deriving the proof). Clearly justify any statement you make, even if the problem doesn't explicitly ask you to. Solving any question flawlessly will add a 4 point bonus for each such solution.
- Good luck!

1. (a) [2] What is the Taylor series expansion of $\log(1+x)$ around $x=0$? For what x does this series converge ?

   (b) [1] Approximate $\int_0^1 \log(1+x^3)dx$ to one decimal place.

   (c) [2] To approximate $\int_0^1 \log(1+x^3)dx$ with an error $\le \epsilon$, how many terms of the power series are enough ? You can give your answer in the $O()$ notation.

   (d) [1] Consider the power series

   $$R(x) = \sum_{n=1}^{\infty} \frac{x^n}{n^2}.$$

   What is the radius of convergence of the given power series ?

   (e) [2] Evaluate the integral $\int_0^{\infty} \log(1+e^{-t})dt$.

2. (a) [3] Find the derivatives of the following matrix functions

   $$f(X) = \sin(\text{Trace}(A^{\mathsf{T}}X)), \quad g(X) = \text{Trace}(A^{\mathsf{T}}\sin(X)),$$

   where $A \in \mathbb{R}^{m\times n}$.

   (b) [4] Let $f : \mathbb{R}^3 \mapsto \mathbb{R}$ be given by $f(x_1, x_2, x_3) = \log\left(e^{x_1} + e^{x_2} + e^{x_3}\right)$.

   i. Compute the Hessian $H(\underline{x})$ of f.

   ii. Show that the Hessian satisfies $\underline{v}^{\mathsf{T}}H(\underline{x})\underline{v} \ge 0$, for any $\underline{v}, \underline{x} \in \mathbb{R}^3$.

   iii. What can you say about $\nabla^2 f = \nabla.(\nabla f)$?

   (c) [2] Which points on the surface $xy^2z^3 = 2$ are closest to the origin?

3. (a) [2] Find the center of mass of a sector of a circle centered at the origin with radius r and central angle θ. Assume a uniform density.

   (b) [2] For the previous part, find the moment of intertia about the origin. You may assume a uniform density.

   (c) [4] Evaluate the following integrals. In each case, $S = \{(x_1, x_2) : x_1, x_2 \ge 0\}$ is the positive quadrant of $\mathbb{R}^2$, and λ_1, λ_2 are positive constants.

   $$\iint_S \lambda_1\lambda_2(x_1+x_2)e^{-\lambda_1 x_1 - \lambda_2 x_2}dA, \quad \iint_S \lambda_1\lambda_2 \min\{x_1, x_2\}e^{-\lambda_1 x_1 - \lambda_2 x_2}dA \text{ and } \iint_S \lambda_1\lambda_2|x_1 - x_2|e^{-\lambda_1 x_1 - \lambda_2 x_2}dA$$

4. (a) [4] Let $\mathbf{S} : \mathbb{R}^2 \mapsto \mathbb{R}^2$ be the spin field $\mathbf{S} = -y\mathbf{i} + x\mathbf{j}$, and $r = \sqrt{x^2+y^2}$ be the distance of a point $(x, y) \in \mathbb{R}^2$ to the origin. The following questions pertain to the field $\mathbf{T} = \mathbf{S}/r^2$.

   i. What is the divergence of $\mathbf{T}$?

   ii. What is the curl of $\mathbf{T}$?

   iii. Find $\iint_D |\nabla \times \mathbf{T}|dA$, where $D \subseteq \mathbb{R}^2$ is a disc of radius R.

   iv. (True or False): We can express $\nabla \times \mathbf{T}$ in terms of the two dimensional impulse δ as $\nabla \times \mathbf{T} = c\delta$. If true, also give the value of c.

(b) [4] Let R be the region in the $x-y$ plane enlcosed by the axes, $x=1$ and $y=x^3$. Note that R has three boundaries: a horizontal segment (call it C_1), a vertical segment (say C_2) and a curved segment (say C_3). Let $\mathbf{F}=(1+y^2)\mathbf{j}$ be a 2-d vector field.

i. Find the flux of $\mathbf{F}$ out of (the boundary of) R using the Green's Theorem.

ii. Find the flux out of C_1 and C_2.

iii. Find the flux out of C_3.

5. [7] Answer each of the following true/false questions (as usual, provide reasoning. If the statement is true, give a proof. If false, provide a counterexample.)

(a) For any vector field $\mathbf{F}$, the curl $\nabla\times\mathbf{F}$ is perpendicular to $\mathbf{F}$.

(b) if $\rho(x,y)$ is a positive function on the plane such that $\iint_{\mathbb{R}^2}\rho\, dA=1$, then $\rho(x,y)\le 1$ for all (x,y)

(c) For a 2-d vector field $\mathbf{F}$, if the divergence $\nabla.\mathbf{F}$ and curl $\nabla\times\mathbf{F}$ are well defined and zero on the whole plane, then $\mathbf{F}=0$.

(d) If $\sum_0^\infty b_n x^n$ has radius of convergence $|x|<2$, then so does $\sum_0^\infty b_n x^{n+1}/(n+1)$.

(e) If $\mathbf{F}$ is a 2-d conservative vector field with potential function f, then $\mathbf{F}$ is perpendicular to the level curve of f.

(f) The gradient of $f(\underline{x})=\underline{x}^\top A\underline{x}$ is $\nabla f(\underline{x})=2A\underline{x}$.

(g) The function $f:\mathbb{R}^{n\times n}\mapsto\mathbb{R}$ defined by $f(X)=\underline{a}^\top X\underline{a}+\underline{b}^\top X\underline{b}-\underline{a}^\top X\underline{b}-\underline{b}^\top X\underline{a}$ has no maxima, minima or saddle points.